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Inventor(s)

ANDERSEN; Tomas Sune

WIRELINE INTERVENTION TOOL STRING

Abstract

The present invention relates to a wireline intervention tool string (1) for operating in a well (2) having a longitudinal axis, a top (51) and a bottom (52), comprising a tool housing (3) having a longitudinal tool axis, a first part connected to a wireline and configured to face the top of the well, the first part comprising an anchoring section (6) and configured to be rotationally stationary during operation, an operational tool (7) configured to rotate relative to the first part during operation, a driving unit (8) arranged in the first part and configured to provide rotational force to the operational tool, the driving unit being powered via the wireline, and a vibration-generating unit (9) for providing vibration force to the operational tool, wherein the wireline intervention tool string further comprises a vibration-dampening unit (10) arranged in a longitudinal position uphole relative to the operational tool for preventing vibrations generated by the vibration-generating unit from releasing tool sections from other tool sections of the wireline intervention tool string, at least in the first part. The invention also relates to downhole tool system comprising the wireline intervention tool string and a driving section/downhole tractor (15).

Inventors: ANDERSEN; Tomas Sune (ZUG, CH)

Applicant: WELLTEC A/S (Allerød Denmark, DK)

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Background/Summary

[0001] The present invention relates to a wireline intervention tool string for operating in a well having a longitudinal axis, a top and a bottom. The invention also relates to a downhole tool system comprising the wireline intervention tool string and a driving section/downhole tractor.

[0002] When intervening a well with a wireline tool, a very limited amount of power, corresponding to that used by a hairdryer, is available 15 kilometres down the well. Machining operations so far down the well can thus be very difficult, if not impossible, when the material to be machined is too hard.

[0003] It is an object of the present invention to wholly or partly overcome the above disadvantages and drawbacks of the prior art. More specifically, it is an object to provide an improved wireline intervention tool string which is able to machine in hard materials far down the well.

[0004] Furthermore, when a wireline intervention tool string is configured to machine in hard materials far down the well, the contact between the tool string and the hard material can mean that when the machining tool comes into contact with the hard material, any vibration created between the tool string and the hard material may be amplified by the stationary hard material, as the hard material may provide a counterforce to the vibration generated in the tool string and thereby cause the vibration to be transmitted into the remaining parts of the tool string.

[0005] The above objects, together with numerous other objects, advantages and features, which will become evident from the below description, are accomplished by a solution in accordance with the present invention by a wireline intervention tool string for operating in a well having a longitudinal axis, a top and a bottom, comprising: [0006] a tool housing having a longitudinal tool axis, [0007] a first part connected to a wireline and configured to face the top of the well, the first part comprising an anchoring section and configured to be rotationally stationary during operation, [0008] an operational tool configured to rotate relative to the first part during operation, [0009] a driving unit, such as an electric motor, arranged in the first part and configured to provide rotational force to the operational tool, the driving unit being powered via the wireline, and [0010] a vibration-generating unit for providing vibration force to the operational tool, wherein the wireline intervention tool string further comprises a vibration-dampening unit arranged in a longitudinal position uphole relative to the operational tool for preventing vibrations generated by the vibration-generating unit from damaging tool sections that are uphole to the operational tool and/or to prevent the vibration generated by the vibration-generating unit from releasing tool sections from other tool sections of the wireline intervention tool string, at least in the first part. [0011] Alternatively, the vibration-dampening unit may protect the functionality of the tool sections that are uphole to the operational tool and to reduce the risk that the vibration of the operational tool reduces the operational functionality of the tool sections. As an example, the vibration-dampening unit may protect the anchor and/or anchoring section from losing their anchoring functionality or loosening the grip to the inner surface of the well tubular structure. [0012] Thus, the vibration-dampening unit may be arranged in a longitudinal position between the vibration-generation unit and/or the operational tool and other tool sections of the wireline

intervention tool. Therefore, the provision of a vibration-dampening unit between the operational tool and the other tool sections the vibration-dampening unit may protect the other tool sections from the vibration of the vibration-generating unit and/or the operational tool and ensure that at least a majority of the vibration created by the vibration-generation tool is dampened or isolated from the other tool sections, and thereby minimizing the risk that the vibration may separate other tool sections or that the vibration may cause damage to the other tool sections.

[0013] Thus, a majority of the vibration generated by the vibration-generation unit or the vibration of the operational tool will not be transferred to the remaining sections of the tool string and the risk of vibration damage to the tool string or the risk of inadvertent release of tool sections from each other will be reduced.

[0014] Also, the vibration-dampening unit may be a vibration-absorption or vibration-isolation unit.

[0015] Furthermore, the wireline intervention tool string may comprise a second part configured to face the bottom of the well, and the operational tool may be arranged in the second part of the wireline intervention tool string, the operational tool comprising a bit for machining, such as grinding and/or milling.

[0016] In addition, the vibration-generating unit may provide oscillating vibration force at a frequency of more than 50 Hz, or more preferably of more than 100 Hz, or even more preferably of more than 200 Hz.

[0017] Moreover, the vibration-generating unit may provide oscillating vibration force at a frequency of less than 800 Hz, or more preferably less than 600 Hz, or even more preferably less than 450 Hz.

[0018] Further, the vibration-dampening unit may be arranged in a longitudinal position between the operational tool and the first part.

[0019] Furthermore, the vibration-dampening unit may be positioned in a longitudinal position that is between the operational tool and the anchoring section.

[0020] Thus, by arranging the vibration-dampening unit between the operational tool and the first part and/or the anchoring section, the operational tool may in operation be arranged between a first stationary part (anchoring section/first part) and a stationary part of the well, i.e. a hard material to be machined. Thus, it may be ensured that any vibration that is transmitted from the hard material via the operational tool is absorbed by the vibration-dampening unit or is isolated or dampened prior to it being transferred to the first part and/or the anchoring section.

[0021] Thus, there is a reduced risk that the vibration of the operational tool and/or the vibration-generating unit will affect the anchoring of the tool during the downhole operation.

[0022] Furthermore, the vibration-dampening unit may be arranged in a longitudinal position between the driving unit and the operational tool. Thus, the vibration dampening unit may protect the driving unit from the vibration, and the risk of damage to the driving unit during a downhole operation is reduced.

[0023] Also, the vibration-dampening unit may be configured to be stationary during operation.

[0024] Furthermore, the vibration-dampening unit may be configured to rotate relative to the first part during operation.

[0025] In addition, the driving unit may be configured to provide rotational force to the vibration-generating unit, where the vibration-generating unit is configured to transform the rotational force into the vibration force in the longitudinal tool axis.

[0026] Moreover, the wireline intervention tool string may further comprise a second driving unit that is configured to provide rotational force to the vibration-generating unit, where the vibration-generating unit is configured to transform the rotational force into the vibration force.

[0027] Further, the driving unit may be connected with the operational tool via a drive shaft.

[0028] Also, the tool string may further comprise a gearing unit arranged in a longitudinal position between the driving unit and the operational tool and/or between the driving unit and the vibration-

generating unit.

[0029] Furthermore, the gearing unit may be connected with the driving unit via a drive shaft and may be connected with the operational tool and/or the vibration-generating unit via a second drive shaft and/or a third drive shaft.

[0030] In addition, the gearing unit may be configured to provide a first rotational speed to the operational tool and/or a second rotational speed to the vibration-generating unit, where the first rotational speed is different from the second rotational speed.

[0031] Moreover, the gearing unit may provide a gearing ratio of 1:1 to the operational tool, but may have a different gearing ratio to the vibration-generating unit, or vice versa, i.e. one of the parts may rotate at the same speed as the drive shaft.

[0032] Further, the first rotational speed may be lower than the second rotational speed.

[0033] Also, the tool string and/or the first part may further comprise one or more compensating units, motors, gears, electrical controls or pumps.

[0034] Furthermore, the wireline intervention tool string may also comprise a slip ring.

[0035] In addition, the anchoring section may be configured to anchor the first part to the well to prevent rotational movement of the first part.

[0036] Moreover, the anchoring section may be a driving section such as a self-propelling tool section/downhole tractor having wheels on arms.

[0037] Furthermore, the vibration-dampening unit may be arranged in a longitudinal position between the driving section and the operational tool. Thus, the vibration dampening unit may protect the driving unit from the vibration, and the risk of damage to the driving unit during a downhole operation is reduced.

[0038] Further, the anchoring section may comprise projectable parts for anchoring the first part in the well.

[0039] Also, the vibration-generating unit may be arranged in the first part.

[0040] Furthermore, the vibration-generating unit may provide oscillating vibration force in a direction that is parallel, coaxial and/or coincident to the longitudinal tool axis of the wireline intervention tool string and/or the longitudinal axis of the well.

[0041] In addition, the vibration-generating unit may be arranged in the second part.

[0042] Moreover, the vibration-dampening unit may be arranged in the second part.

[0043] Further, the vibration-dampening unit may be arranged in a longitudinal position between the vibration-generating unit and the first part.

[0044] Also, the vibration-generating unit may be arranged in a longitudinal position between the vibration-dampening unit and the operational tool.

[0045] Furthermore, the vibration-generating unit may comprise a wave bearing or a magnetostrictive oscillator providing the vibration force.

[0046] In addition, the vibration-generating unit may comprise a vibration-transmission unit comprising a spring system.

[0047] Also, the vibration-transmission unit may be configured to transmit and enhance the vibration force generated by the vibration-generating unit.

[0048] Moreover, the spring system may be formed as a monolithic whole and shaped as stacked Belleville Washers.

[0049] Further, the spring system may comprise two or more frusto-conical springs arranged in series and/or two or more Belleville Springs arranged in series.

[0050] Also, the vibration-generating unit may comprise a rotatable cam mechanism comprising a first cam profile rotated in relation to a cooperating second cam profile.

[0051] Finally, the invention relates to a downhole tool system comprising the wireline intervention tool string and a driving section/downhole tractor.

Description

[0052] The invention and its many advantages will be described in more detail below with reference to the accompanying schematic drawings, which for the purpose of illustration show some non-limiting embodiments and in which:

[0053] FIG. 1 shows a wireline intervention tool string seen from the side,

[0054] FIG. 2 shows another wireline intervention tool string seen from the side, and

[0055] FIG. 3 shows yet another wireline intervention tool string seen from the side.

[0056] All the figures are highly schematic and not necessarily to scale, and they show only those parts which are necessary in order to elucidate the invention, other parts being omitted or merely suggested.

[0057] FIG. 1 shows a wireline intervention tool string **1** operating in a well **2** having a longitudinal axis **L**, a top **51** and a bottom **52**. The wireline intervention tool string **1** comprises a tool housing **3** having a longitudinal tool axis **LT** and a first part **4** connected to a wireline **5** and configured to face the top of the well **2**. The first part **4** comprises an anchoring section **6** for maintaining the first part **4** rotationally stationary during an operation. The wireline intervention tool string **1** further comprises an operational tool **7** configured to rotate relative to the first part **4** during operation and a driving unit **8**, such as an electric motor, arranged in the first part **4** and configured to provide rotational force to the operational tool **7**. The driving unit **8** is powered via the wireline **5**. The wireline intervention tool string **1** further comprises a vibration-generating unit **9** for providing vibration force to the operational tool **7** to enhance the machining operation of the operational tool **7**. The operational tool **7** may be used to bore out a stuck valve partly blocking the passage of hydrocarbon-containing fluid in the well tubular metal structure or to machine in and widen a restriction in the well tubular metal structure, or to clean the inner face of the well tubular metal structure. The vibration-generating unit **9** is connected with the operational tool **7** via a drive shaft **16** rotating the operational tool **7**. The vibration-generating unit **9** provides oscillating vibration force in a direction that is parallel, coaxial and/or coincident to the longitudinal tool axis of the wireline intervention tool string and/or the longitudinal axis of the well **2**.

[0058] By having vibrations in the form of an oscillating vibration force in a direction that is parallel, coaxial and/or coincident to the longitudinal tool axis, the operational tool **7** is able to machine in harder materials further down the well than without such vibration force. Consequently, operations which were not possible with prior art tools are now feasible and easily done in one run and by means of a wireline and thus without the need for heavy equipment, such as coiled tubing on a rig.

[0059] The wireline intervention tool string **1** having tool sections further comprises a vibration-dampening unit **10** arranged in a longitudinal position uphole relative to the operational tool **7** for preventing vibrations generated by the vibration-generating unit **9** from releasing tool sections from other tool sections, at least in the first part. The vibration-dampening unit **10** is arranged in a longitudinal position between the rotationally operational tool and the stationary first part **4**. The vibration-generating unit **9** provides oscillating vibration force at a frequency of preferably more than 200 Hz, and if such vibrations are allowed to spread in the wireline intervention tool string **1** and in particular to the stationary first part **4**, some of the tool sections are at risk of releasing themselves from other tool sections, jeopardising the operation and creating a significant risk that these tool sections will be lost in the well **2** and difficult to retrieve from the well **2** again. Thus, the vibration-dampening unit **10** may be a vibration-absorption or vibration-isolation unit, depending on the design.

[0060] As can be seen in FIG. 1, the wireline intervention tool string **1** comprises a second part **12** configured to face the bottom of the well **2**, and the operational tool **7** is arranged in the second part **12** of the wireline intervention tool string **1**. The operational tool **7** comprises a bit **14** for

machining, such as grinding and/or milling, in an obstacle or object in front of the operational tool 7.

[0061] In FIG. 1, the vibration-dampening unit 10 is configured to rotate relative to the first part 4 during operation and thus to rotate along with at least part of a second part 12 of the wireline intervention tool string 1. The vibration-generating unit 9 is arranged in the second part 12 and rotates. The anchoring section 6 comprises projectable parts 28 for anchoring the first part 4 in the well 2. The wireline intervention tool string 1 further comprises a second driving unit 15 that is configured to provide rotational force to the vibration-generating unit 9, where the vibration-generating unit 9 is configured to transform the rotational force into the vibration force.

[0062] In FIG. 2, the vibration-dampening unit 10 is configured to be stationary during operation, and the anchoring section 6 is configured to anchor the first part 4 to the well 2 to prevent rotational movement of the first part 4. The vibration-dampening unit 10 is arranged in a longitudinal position between the vibration-generating unit 9 and the first part 4. The anchoring section 6 is a driving section 25, such as a self-propelling tool section, i.e. a downhole tractor having wheels 26 on arms 27. The driving unit 8 is configured to provide rotational force to the vibration-generating unit 9, where the vibration-generating unit 9 is configured to transform the rotational force into the vibration force in the longitudinal tool axis. The wireline intervention tool string 1 further comprises a gearing unit 17 arranged in a longitudinal position between the driving unit 8 and the operational tool 7, and even between the driving unit 8 and the vibration-generating unit 9. The gearing unit 17 is connected with the driving unit 8 via the drive shaft 16, and the gearing unit 17 is connected with the operational tool 7 and the vibration-generating unit 9 via a second drive shaft 18 and a third drive shaft 19. The gearing unit 17 is configured to provide a first rotational speed to the operational tool 7 and/or a second rotational speed to the vibration-generating unit 9, where the first rotational speed is different from the second rotational speed. The gearing unit 17 may provide a gearing ratio of 1:1 to the operational tool 7, but may have a different gearing ratio to the vibration-generating unit 9, or vice versa, i.e. one of the parts may rotate at the same speed as the drive shaft 16, and the other part at a higher speed. The first rotational speed in FIG. 2 is lower than the second rotational speed.

[0063] The first part of the wireline intervention tool string 1 of FIG. 1 comprises an electrical control 23 connected to the wireline 5, a compensating unit 20 for maintaining a predetermined pressure in the wireline intervention tool string 1, a motor 21 powered by the wireline 5 and a pump 24 driven by the motor 21. The wireline intervention tool string 1 further comprises a slip ring 36 to transmit power to the electric motor 15, even though the motor 15 rotates in relation to the first part 4 and the wireline 5.

[0064] In FIG. 3, the vibration-generating unit 9 is arranged in the stationary first part 4 so that it is only the operational tool 7 that rotates. The drive shaft 16 rotates the operational tool 7, and the second drive shaft 18 is for driving the vibration-generating unit 9.

[0065] The vibration-generating unit 9 comprises a wave bearing or a magnetostrictive oscillator providing the vibration force. In another aspect, the vibration-generating unit 9 may comprise a rotatable cam mechanism comprising a first cam profile rotated in relation to a cooperating second cam profile. The vibration-generating unit 9 may comprise a vibration-transmission unit 30 comprising a spring system configured to transmit and enhance the vibrations generated by the vibration-generating unit 9. The spring system may be formed as a monolithic whole and shaped as stacked Belleville Washers. The spring system may also comprise two or more frusto-conical springs arranged in series and/or two or more Belleville Springs arranged in series.

[0066] The wireline intervention tool string 1 may further comprise a stroking tool, which is a tool providing an axial force on the operational tool 7. The stroking tool comprises an electric motor for driving a pump. The pump pumps fluid into a piston housing to move a piston acting therein. The piston is arranged on the stroker shaft. The pump may pump fluid out of the piston housing on one side and simultaneously suck fluid in on the other side of the piston.

[0067] By “fluid” or “well fluid” is meant any kind of fluid that may be present in oil or gas wells downhole, such as natural gas, oil, oil mud, crude oil, water, etc. By “gas” is meant any kind of gas composition present in a well, completion or open hole, and by “oil” is meant any kind of oil composition, such as crude oil, an oil-containing fluid, etc. Gas, oil and water fluids may thus all comprise other elements or substances than gas, oil and/or water, respectively.

[0068] By “casing” or “well tubular metal structure” is meant any kind of pipe, tubing, tubular, liner, string, etc., used downhole in relation to oil or natural gas production.

[0069] In the event that the tool is not submergible all the way into the casing, a downhole tractor can be used to push the tool all the way into position in the well. The downhole tractor may have projectable arms having wheels, wherein the wheels contact the inner surface of the casing for propelling the tractor and the tool forward in the casing. A downhole tractor is any kind of driving tool capable of pushing or pulling tools in a well downhole, such as a Well Tractor®.

[0070] Although the invention has been described above in connection with preferred embodiments of the invention, it will be evident to a person skilled in the art that several modifications are conceivable without departing from the invention as defined by the following claims.

Claims

1.-15. (canceled)

16. A wireline intervention tool string for operating in a well having a longitudinal axis, a top and a bottom, comprising: a tool housing having a longitudinal tool axis, a first part connected to a wireline and configured to face the top of the well, the first part comprising an anchoring section and being configured to be rotationally stationary during operation, an operational tool configured to rotate relative to the first part during operation, a driving unit arranged in the first part and configured to provide rotational force to the operational tool, the driving unit being powered via the wireline, and a vibration-generating unit for providing vibration force to the operational tool, wherein the wireline intervention tool string further comprises a vibration-dampening unit arranged in a longitudinal position uphole relative to the operational tool for preventing vibrations generated by the vibration-generating unit from damaging tool sections that are uphole to the operational tool and/or to prevent the vibration generated by the vibration-generating unit from releasing tool sections from other tool sections of the wireline intervention tool string, at least in the first part.

17. A wireline intervention tool string according to claim 16, wherein the wireline intervention tool string comprises a second part configured to face the bottom of the well, and the operational tool is arranged in the second part of the wireline intervention tool string, the operational tool comprising a bit for machining, such as grinding and/or milling.

18. A wireline intervention tool string according to claim 16, wherein the vibration-generating unit provides oscillating vibration force at a frequency of more than 50 Hz, or more preferably of more than 100 Hz, or even more preferably of more than 200 Hz.

19. A wireline intervention tool string according to claim 16, wherein the vibration-dampening unit is arranged in a longitudinal position between the operational tool and the first part.

20. A wireline intervention tool string according to claim 16, wherein the vibration-dampening unit is configured to be stationary during operation.

21. A wireline intervention tool string according to claim 16, wherein the vibration-dampening unit is configured to rotate relative to the first part during operation.

22. A wireline intervention tool string according to claim 16, wherein the driving unit is configured to provide rotational force to the vibration-generating unit, where the vibration-generating unit is configured to transform the rotational force into the vibration force in the longitudinal tool axis.

23. A wireline intervention tool string according to claim 16, further comprising a second driving unit that is configured to provide rotational force to the vibration-generating unit, where the vibration-generating unit is configured to transform the rotational force into the vibration force.

- 24.** A wireline intervention tool string according to claim 16, wherein the wireline intervention tool string further comprises a gearing unit arranged in a longitudinal position between the driving unit and the operational tool and/or between the driving unit and the vibration-generating unit.
- 25.** A wireline intervention tool string according to claim 24, wherein the gearing unit is configured to provide a first rotational speed to the operational tool and/or a second rotational speed to the vibration-generating unit, where the first rotational speed is different from the second rotational speed.
- 26.** A wireline intervention tool string according to claim 16, wherein the vibration-generating unit provides oscillating vibration force in a direction that is parallel, coaxial and/or coincident to the longitudinal tool axis of the wireline intervention tool string and/or the longitudinal axis of the well.
- 27.** A wireline intervention tool string according to claim 16, wherein the vibration-generating unit is arranged in a longitudinal position between the vibration-dampening unit and the operational tool.
- 28.** A wireline intervention tool string according to claim 16, wherein the vibration-generating unit comprises a wave bearing or a magnetostrictive oscillator providing the vibration force.
- 29.** A wireline intervention tool string according to claim 16, wherein the vibration-generating unit comprises a vibration-transmission unit **(30)** comprising a spring system.
- 30.** A downhole tool system comprising the wireline intervention tool string according to claim 16 and a driving section/downhole tractor.
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